Stefan Herytash

sherytash.github.io

DRAFT

Research Statement

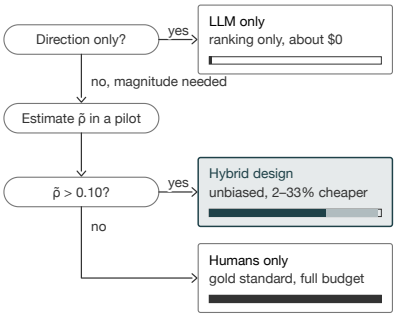
Overview

A respondent's answer to a questionnaire used to cost time, and a manuscript or a review cost its author weeks. Part of what made each of them informative was that cost. Language models have made all of them cheap. My research follows two consequences for marketing. The first concerns measurement: what a marketing researcher can still learn from answers a model can produce, and what can be measured now that simulated answers cost almost nothing. The second concerns certification: the journals and platforms that vouch for a manuscript or a wage figure have to be rebuilt once the signal they relied on is cheap.

Measurement

I trained in mathematics at TU Wien and the University of Zurich. My PhD at LMU Munich's Institute for Market-Based Management studied how far large language models can simulate human decisions, and I treat simulation as a marketing research method that has to pass the tests any new instrument faces. In a solo paper under revision at the *International Journal of Research in Marketing*, I simulate 70 treatment effects from 30 published marketing studies with an open-weights language model, at the level of the instrument: each synthetic respondent completes the original questionnaire in one randomized condition. Simulated and human effects correlate at $r = .73$, but the calibration slope is .38, so the simulations understate effect sizes. They can rank ideas before a test is run; they cannot yet price them.

The paper turns this into a decision rule (right). If a study needs only the direction of an effect, the simulation alone gives a ranking at almost no cost. If it needs the size, a paired pilot estimates the correlation between human and simulated responses. Above 0.10, a hybrid design corrects the simulation with a human sample, stays unbiased and, in the paper's illustration, costs 2 to 33 percent less than humans alone. Below 0.10 the study runs on humans.



A companion paper bounds that saving. The share of human respondents a study still needs falls as synthetic respondents are added, but it stops at $1 - \rho^2$, where ρ is the correlation within a study between a human's response and the simulator's response for that human. The floor holds however large the synthetic sample: at $\rho = 0.3$ the saving is 9 percent, and halving the human sample takes ρ of at least 0.71.

Which model to simulate with is itself a measurement question. Psych-201, a dataset I contributed to as one of 79 authors, measures behavioural alignment at scale. Post-training, the stage that turns a base model into an assistant, reduces alignment with human behaviour across model families and sizes, and conditioning a model on a participant's own information does not improve predictions for individuals.

The econometric side of the same question is the many-analyst study forthcoming in the *International Journal of Research in Marketing*, in which 156 independent teams, mine among them, estimated brand price elasticities from the same data. Across 10,415 estimates the team averages range from −7.3 to +3.1, and choices on time aggregation, fixed effects, time controls and marketing-mix covariates explain the largest share of that spread. Pricing and promotion planning rest on such estimates, so the analyst's choices belong to the measurement instrument as much as the questionnaire does.

Certification

When manuscripts cost almost nothing to write, a journal's accept verdict carries less of the information it used to. *After the Fax Machine Burns*, a solo paper submitted to *Management Science*, develops a system a publisher could run: every submission is published on arrival, and an open marketplace produces a public, revisable grade afterward. Modelling the five parties a publication system serves, the paper shows that authors, committees and readers weakly prefer a grade from the publisher's own panels to the accept verdict.

Platforms and professions certify too. *Threshold Migration to the Patient*, accepted at ANZAM 2026, follows a consumer device into a profession's gate: a wearable's algorithm now initiates triage while the clinician keeps disposition. In the labour market, kununu, the largest employer-review and salary platform in the German-speaking countries, shows a salary band only once enough employees have reported one. Its display threshold decides what counts publicly as a normal wage, and a regression discontinuity at that threshold can show what it does to applicant flows. A second paper, *The Matching Cost of Pay Transparency*, models how firms respond to pay-transparency regulation and what follows for labour-market outcomes; the formal model is complete and the empirical estimation is in progress.

Next

SimuLabs, the engine behind the simulation papers, is open source (Rust). It runs a study from its original survey files, every statistic in the benchmark paper regenerates from one deterministic pipeline, and its prediction-powered rectifier returns an unbiased estimate whatever the simulator does. I am turning the prototype into a tool that marketing researchers and students can run on their own questionnaires. For the salary work, a data partnership with kununu is in negotiation.